KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4541 | 4 Credits: 4 | Program Core | Source-grounded

Power Electronics

- To introduce students to the basic theory of power semiconductor devices and

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4541 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4541.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Electrical Engineering
Course code	4541
Course title	Power Electronics
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To introduce students to the basic theory of power semiconductor devices and
- passive components, and their practical applications in power electronics.
- ● To familiarize students with the principle of operation of different power electronic
- converters and their applications.
- ● To provide a strong foundation for further study of power electronics

Course outcomes

CO	Official outcome	Study level
CO1	a particular application. 16 Understanding Summarize the principle and operation of controlled	See official syllabus
CO2	14 Understanding rectifiers. Summarize the principle and operation of DC and AC Understanding	See official syllabus
CO3	converters. 12	See official syllabus
CO4	Summarize various inverters and electric drives. 16 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2

Row	Extracted official mapping
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	application.	4	As specified
Module 2	Summarize the principle and operation of controlled rectifiers.	4	As specified
Module 3	Summarize the principle and operation of DC and AC converters.	3	As specified
Module 4	Summarize various inverters and electric drives	4	As specified

MODULE 1

application.

This module is presented from the official syllabus scope for Power Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: application.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

devices.

devices. is an official topic in Module 1 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify devices. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, devices. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What devices. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ devices. □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off

Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off is an official topic in Module 1 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the operation of scr 3 understanding classify and explain turn on and turn off should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding methods of SCR.

4 Understanding methods of SCR. is an official topic in Module 1 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding methods of SCR. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding methods of scr. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding methods of SCR. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding methods of SCR. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols (MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only)

Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols (MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods - SCR Turn-On methods -list various methods - basic concepts - gate triggering - forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only) is an official topic in Module 1 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols (MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only) using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the idea of datasheets of scr 1 understanding power electronic devices- list various devices -symbols (mosfet, igbt,gto, lascr, scs, ujt, diac, triac and scr) -mosfet - list different types -structure and working of n-channel enhancement type-igbt-- structure-working principle - applications-ujt- schematic diagram and working- diac & triac-schematic diagram- working- v/i characteristics. silicon controlled rectifier (scr)-structure-v-i characteristics- definitions-holding current- latching current scr turn on and turn off methods -scr turn-on methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-scr turn-off methods- list various methods- basic concepts-thyristor protection circuits-snubber circuits. specifications of scr- list parameters (datasheet interpretation only) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols (MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols (MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types - structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC- schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)- structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only) □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
application.
               Summarize the operation of power electronic
    M1.01      devices.                                            8
Understanding

    M1.02      Explain the operation of SCR                        3
Understanding
               Classify and explain turn on and turn off
    M1.03                                                          4
Understanding
               methods of SCR.

    M1.04      Summarize the idea of datasheets of SCR             1
Understanding

 Contents:

 Power Electronic Devices- List various devices -Symbols ( MOSFET,
 IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list
 different types -structure and working of N-channel enhancement type-IGBT--
 structure-working principle - applications-UJT- schematic diagram and working-
 DIAC & TRIAC-schematic diagram- working- V/I characteristics.

 Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-
holding
 current- latching current
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Summarize the principle and operation of controlled rectifiers.

This module is presented from the official syllabus scope for Power Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the principle and operation of controlled rectifiers.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding wave controlled rectifier Explain the operation of single phase full

4 Understanding wave controlled rectifier Explain the operation of single phase full is an official topic in Module 2 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding wave controlled rectifier Explain the operation of single phase full using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding wave controlled rectifier explain the operation of single phase full should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding wave controlled rectifier Explain the operation of single phase full means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding wave controlled rectifier Explain the operation of single phase full □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

5 Understanding wave controlled rectifier Summarize the operation of single phase

5 Understanding wave controlled rectifier Summarize the operation of single phase is an official topic in Module 2 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding wave controlled rectifier Summarize the operation of single phase using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding wave controlled rectifier summarize the operation of single phase should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding wave controlled rectifier Summarize the operation of single phase means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Understanding wave controlled rectifier Summarize the operation of single phase □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding semi controlled rectifiers Outline the operation of three phase

2 Understanding semi controlled rectifiers Outline the operation of three phase is an official topic in Module 2 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding semi controlled rectifiers Outline the operation of three phase using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding semi controlled rectifiers outline the operation of three phase should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding semi controlled rectifiers Outline the operation of three phase means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 2 Understanding semi controlled rectifiers Outline the operation of three phase [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers (listing only) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only)

3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers (listing only) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only) is an official topic in Module 2 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers (listing only) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only) using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding controlled rectifiers seriestest1 1 single phase half wave controlled rectifiers – terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -types of controlled rectifiers (listing only) single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(r load - rl load with and without freewheeling diode) single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with r load ,rl load only, rl load and freewheeling diode -continuous mode only) single phase semi controlled rectifier - circuit diagram, waveforms and operation (with rl load) three phase bridge converter with resistive load (basic diagram with concept only) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers (listing only) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers (listing only) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only) □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize the principle and operation of controlled rectifiers.

            Illustrate the operation of single phase half
  M2.01                                                     4
Understanding
            wave controlled rectifier
            Explain the operation of single phase full
  M2.02                                                     5
Understanding
            wave controlled rectifier
            Summarize the operation of single phase
  M2.03                                                     2
Understanding
            semi controlled rectifiers
            Outline the operation of three phase
  M2.04                                                     3
Understanding
            controlled rectifiers
            SeriesTest1                                     1

Contents:
Single phase half wave controlled rectifiers – Terms and definitions – controlled
rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled
rectifiers ( listing only )
Single phase half wave controlled rectifiers-circuit diagram, waveforms and
operation( R load - RL load with and without freewheeling diode)
Single phase full wave controlled rectifiers-circuit      diagram,    waveforms
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarize the principle and operation of DC and AC converters.

This module is presented from the official syllabus scope for Power Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the principle and operation of DC and AC converters.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost

Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost is an official topic in Module 3 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the operation of dc choppers 3 understanding illustrate the classification of dc chopper 5 understanding m3.02 explain the operation of buck and boost should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Understanding converters

2 Understanding converters is an official topic in Module 3 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding converters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding converters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding converters means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 2 Understanding converters [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage -simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant (Type E)-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type)

Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant (Type E)-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type) is an official topic in Module 3 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage -simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant (Type E)-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the operation of cycloconverters 2 understanding dc chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(typea,typeb)-twoquadrant(typec,typed)-fourquadrant (type e)-applications of chopper buck and boost converters-(operation with diagram only) cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant (Type E)-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations-circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant (Type E)-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type) □□□□□□□□□□□ □□□□□

definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize the principle and operation of DC and AC converters.

  M3.01      Summarize the operation of DC choppers          3
Understanding

             Illustrate the classification of DC chopper     5
Understanding
  M3.02
             Explain the operation of buck and boost
  M3.03                                                      2
Understanding
             converters

  M3.04      Summarize the operation of Cycloconverters      2
Understanding
 Contents:
 DC Chopper-basic principle of operation with waveforms- equation of output
 voltage - simple problems- control strategies- constant frequency control -
 variable frequency control
 Types of choppers -concept of quadrant operations- circuit diagram and basic
 concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)-
 fourquadrant ( Type E )-Applications of chopper

 Buck and boost converters-(operation with diagram only)
 Cycloconverters –types –single phase to single phase step up cycloconverter
(midpoint
 type)
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Summarize various inverters and electric drives

This module is presented from the official syllabus scope for Power Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize various inverters and electric drives
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the operation of basic inverters

Explain the operation of basic inverters is an official topic in Module 4 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the operation of basic inverters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the operation of basic inverters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the operation of basic inverters means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain the operation of basic inverters [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate the operation of bridge inverters

Illustrate the operation of bridge inverters is an official topic in Module 4 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the operation of bridge inverters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the operation of bridge inverters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the operation of bridge inverters means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate the operation of bridge inverters □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Explain different voltage control methods

Explain different voltage control methods is an official topic in Module 4 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different voltage control methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different voltage control methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different voltage control methods means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain different voltage control methods □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspowerelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810 1 1038https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1 08108077 2 www.electrical4u.com/electrical-engineering-articles/power-electronics 3 www.swayam.gov.in 4 www.youtube.com 5 wikipedia.org 6. IEEE papers/journals on power electronics transactions

Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,

(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspowerelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810 1 1038https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1 08108077 2 www.electrical4u.com/electrical-engineering-articles/power-electronics 3 www.swayam.gov.in 4 www.youtube.com 5 wikipedia.org 6. IEEE papers/journals on power electronics transactions is an official topic in Module 4 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3

9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspowerelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810 1 1038https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1 08108077 2 www.electrical4u.com/electrical-engineering-articles/power-electronics 3 www.swayam.gov.in 4 www.youtube.com 5 wikipedia.org 6. IEEE papers/journals on power electronics transactions using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify various electric drives. 2 understanding inverter- requirements of a practical inverter- classification- circuit diagram and working - series inverter- parallel inverter- concept of csi and vsi bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for rl load). three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for r loads only. voltage control methods - principle of pulse width modulation- different voltage control methods-single pwm - sinusoidal pwm electric drives-introduction-types-advantages-basic block diagram text/reference: t/r book title/author p.s.bimbhra, power electronics, khanna publications,co.,new delhi(isbn: t1 9788174092151). m.d.singhand khanchandani k.b, power electronics.secondedition, mcgraw t2 hill publishing co. ltd, new delhi,2008, (isbn:9780070583894) muhammedhrashid, power electronics circuits devices and applications, r1 pearson education india,noida,2014.(isbn: 9788131702468) jainalok, power electronics and its applications, penram international r2 publishing,mumbai,2006 (isbn:9788187972136) dr.brgupta,singhal,powerelectronicskkataria&sons,2009(isbn: r3 9788185749532) senp.c.,power electronics,mcgraw-hillpublishingcompanylimited,new delhi. r4 isbn:9780074624005 v.r.moorthyoxfordpublisherspowerelectronicsdevicecircuitsandindustrial r5 applications online resources sl.no website link nptel>>courses>>electrical engineering>>power electronics https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810 1 1038https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1 08108077 2 www.electrical4u.com/electrical-engineering-articles/power-electronics 3 www.swayam.gov.in 4 www.youtube.com 5 wikipedia.org 6. ieee papers/journals on power electronics transactions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspowerelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810 1 1038https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1 08108077 2 www.electrical4u.com/electrical-engineering-articles/power-electronics 3 www.swayam.gov.in 4 www.youtube.com 5 wikipedia.org 6. IEEE papers/journals on power electronics transactions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136)

Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspowerelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810 1 1038https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1 08108077 2 www.electrical4u.com/electrical-engineering-articles/power-electronics 3 www.swayam.gov.in 4 www.youtube.com 5 wikipedia.org 6. IEEE papers/journals on power electronics transactions □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize various inverters and electric drives

   M4.01      Explain the operation of basic inverters             5
Understanding

   M4.02      Illustrate the operation of bridge inverters         7
Understanding

   M4.03     Explain different voltage control methods             2
Understanding

   M4.04      Classify various electric drives.                    2
Understanding

              Series Test II                                       1

 Contents:
 Inverter- requirements of a practical inverter- Classification- circuit diagram
and working -
 series inverter- parallel inverter- concept of CSI and VSI
 Bridge inverters - concept of feedback diodes- single phase bridge inverter -
 half bridge - full bridge (circuit diagram- working wave forms for RL load).
Three
 phase bridge inverter- 180 degree conduction mode- circuit diagram with
 waveforms for R loads only.
 Voltage control methods - Principle of pulse width modulation- different
 voltage control methods-single PWM - sinusoidal PWM
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain devices.. (3 marks)

Answer: Begin with the standard definition or identity of *devices.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding methods of SCR.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding methods of SCR.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols (MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types - structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current-latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering-dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only). (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols (MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list *different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts -gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods-basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only)*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Understanding wave controlled rectifier Explain the operation of single phase full. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding wave controlled rectifier Explain the operation of single phase full*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding wave controlled rectifier Summarize the operation of single phase. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding wave controlled rectifier Summarize the operation of single phase*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding semi controlled rectifiers Outline the operation of three phase. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding semi controlled rectifiers Outline the operation of three phase*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers (listing only) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only). (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers (listing only) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only)*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding converters. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding converters*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms-equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant (Type E)-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type). (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers- singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant (Type E)- Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type)*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain Explain the operation of basic inverters. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the operation of basic inverters*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Illustrate the operation of bridge inverters. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the operation of bridge inverters*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain different voltage control methods. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different voltage control methods*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspowerelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810 1 1038https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1 08108077 2 www.electrical4u.com/electrical-engineering-articles/power-electronics 3 www.swayam.gov.in 4 www.youtube.com 5 wikipedia.org 6. IEEE papers/journals on power electronics transactions. (3 marks)

Answer: Begin with the standard definition or identity of *Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages- basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014. (ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspowerelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810 1 1038https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1 08108077 2 www.electrical4u.com/electrical-engineering-articles/power-electronics 3 www.swayam.gov.in 4 www.youtube.com 5 wikipedia.org 6. IEEE papers/journals on power electronics transactions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **devices.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding wave controlled rectifier Explain the operation of single phase full**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain the operation of basic inverters**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- NPTEL>>Courses>>Electrical Engineering>>Power Electronics
- https://nptel.ac.in/courses/108102145https://nptel.ac.in/courses/10810
- https://nptel.ac.in/courses/108101126https://nptel.ac.in/courses/1
- www.electrical4u.com/electrical-engineering-articles/power-electronics www.swayam.gov.in www.youtube.com wikipedia.org
- IEEE papers/journals on power electronics transactions

IN-PAGE SEARCH

Search this lesson